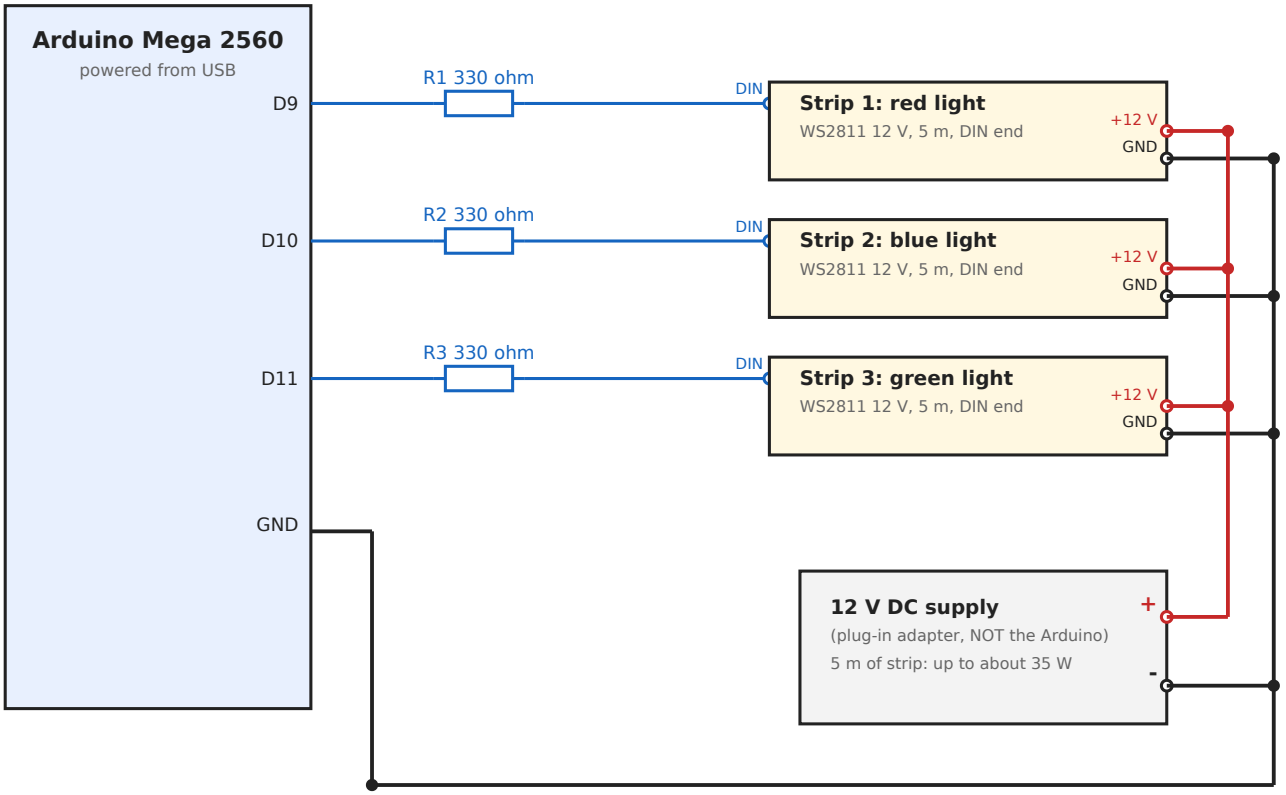
LED strip knight rider - wiring (Arduino Mega 2560 + three 12 V WS2811 strips)



Connections

1. Arduino D9, D10, D11 -> a 330 ohm resistor each -> DIN of strip 1, 2, 3 (the end where the printed arrows point away from). One strip is enough: connect it to D9 (red, slow), D10 (blue) or D11 (green, fast).
2. The +12 V of every strip comes from the 12 V supply, NEVER from an Arduino pin. No 12 V on any Arduino pin.
3. GND of the 12 V supply, GND of all strips and GND of the Arduino are connected together (common ground).
4. The Arduino is powered from USB (the Serial Monitor, 115200 baud, shows one DBG line per second).
5. Optional: a 1000 uF capacitor rated 25 V or more across +12 V and GND at the strip start (plus to +12 V).
6. The built-in LED (pin 13) blinks while the program runs.
7. Brightness is limited to 80 of 255 in the sketch (TEST_LEVEL); one lit pixel per strip needs very little current.